

3. Proposed Outline of Research Plan and Methodology

3.1. Background and Motivation

The adoption of electric vehicles (EVs) is crucial for advancing ESG goals by reducing environmental impact and fostering renewable energy integration. The global EV penetration rate has been steadily increasing. In 2024, 17.1 million EVs were sold worldwide, representing a 25% growth compared to 2023. In China, according to the China Passenger Car Association, 11 million new energy vehicles were sold in 2024, marking a year-on-year increase of 40.7%. EVs accounted for nearly half (47.6%) of all retail car sales in the country. As of the end of 2024, the total number of EVs in Hong Kong surpassed 110,000. Notably, EVs accounted for about 70% of newly registered private vehicles in Hong Kong, a significant rise from just 20% in 2022. This remarkable growth positions Hong Kong as one of the regions with the fastest EV penetration rate.

The rapid adoption of EVs has driven the widespread installation of charging infrastructure. However, range anxiety—the fear of EV battery depleting before reaching a charging station or destination—remains a significant barrier to broader EV adoption. Most EVs require drivers to plug in their vehicles for battery charging, with charging times varying from 30 minutes to several hours, depending on the type of charger and the battery size. While this may not be a major concern for private EV owners with home wall chargers, it presents challenges for electric taxi drivers and ride-sharing operators who rely on quick turnarounds to maintain efficiency. To address this challenge, new fast-charging technologies have been developed. For instance, Tesla Superchargers deliver up to 350 kW of power, enabling vehicles to gain 200 miles of range in just 15 to 30 minutes. However, this still takes considerably longer than refueling a gasoline vehicle. Moreover, the high power demand of superchargers can incur higher demand variations and put significant pressure on the power grid, often necessitating infrastructure upgrades. Unfortunately, such upgrades are not always feasible, particularly in older cities with aging and constrained systems.

Battery swapping stations (BSS) offer an alternative to traditional plug-in charging, where depleted batteries are replaced with fully charged ones. This approach provides several benefits for EV owners, battery manufacturers, and infrastructure planners: **(i) Fast refueling:** battery swapping can take as little as 1-3 minutes, making it comparable to refueling a gasoline car and convenient for drivers. Because of the quick turnaround, BSS usually requires much less space compared to plug-in charging stations which require extensive parking space. This is especially valuable for compact cities such as Hong Kong and New York; **(ii) Smart charging and energy storage:** The charging of depleted batteries are centrally managed. Hence, BSS can charge batteries during off-peak hours, reducing strain on the grid during periods of high electricity demand. Swapping stations can act as energy storage hubs, supporting grid stability by discharging stored energy during peak times; **(iii) Renewable energy integration:** BSS can store excess renewable energy (e.g., from solar or wind) in batteries, aligning EV charging with sustainable energy goals; **(iv) Centralized battery recycling:** BSS can centralize battery collection, making it easier to manage end-of-life batteries for recycling or repurposing. BSS can track battery health and ensure batteries are replaced or repurposed before failure, optimizing resource use.

Hence, battery swapping, as an alternative for EV refueling, is especially suitable for big cities with dense populations. In 2023, the Chinese government placed the development of battery swapping technology and infrastructure as a strategic focus. Nio, a Chinese electric car manufacturer, has made battery swapping a key feature of its business model, installing over 3,000 swap stations across China by the end of 2024. Meanwhile, CATL, the world's top battery maker, announced plans to open 1,000 swap stations in 2025, including locations in Hong Kong and Macao. Robin Zeng, the founder and CEO of CATL, predicted that by 2030, the EV charging market will be evenly divided among battery swapping, home charging piles, and public charging stations.

Despite the great potential of battery swapping, the mass adoption is expected to introduce unparalleled challenges, including, among others, the lack of battery standardization, the large upfront investment, the complex charging and pricing decisions, and the coordination of BSS and power grid. Following consultations with our industry partners, Nio and CATL, we have identified the following key operational challenges that we aim to investigate and address in this project.

- **First**, charging decisions at BSS are challenging due to the time-varying and stochastic nature of swapping demand and the inherent uncertainty of renewable energy generation. The electricity pricing structure further complicates charging decisions: most electricity providers in China apply a time-varying pricing model based on energy consumption, whereas regions like Hong Kong (and parts of the U.S. and Europe) typically use flat rates but impose an additional demand charge based on the peak usage in the billing period. Moreover, BSS can serve as energy storage hubs and reduce strain on the grid by discharging stored energy during peak hours. The charging and discharging decisions have to consider these factors mentioned above, which makes it a high-dimensional decision problem.
- **Second**, the large upfront investment of building a swap station makes it challenging to scale it up. For instance, each of Nio's 2nd-generation BSS costs approximately 3 million CNY (about \$400,000), with batteries accounting for one-third of the total cost. Hence, it is critical to optimize the capacity of the BSS by determining the number of batteries and the size of solar panels to meet predefined service levels while minimize the upfront cost.
- **Third**, battery swapping is a relatively new business model. The pricing strategy of the battery swapping service and its impact is to be explored. Nio has implemented different payment structures including fee for each swap and fee for the added miles. However, the implications of the pricing schemes on consumers' charging choices, the profitability of BSS, the adoption rate of EVs, and the environmental impact remain unclear. Furthermore, the intricate structure of electricity tariffs by the power grid, including time-varying energy cost and demand charge, raises fairness concerns for pricing battery-swapping services, which need to be addressed.
- **Fourth**, it is expected that 602,000 tons of automotive batteries will be retired in China by 2025. The operations of battery swapping will inevitably require more batteries. Recycling such a large volume of retired batteries and minimizing the environmental impact will be a challenging task. BSS can centralize battery collection, making it easier to manage end-of-life batteries for recycling or repurposing. BSS can track battery health and ensure batteries are replaced or repurposed before failure, optimizing resource use. However, the battery recycling process is not straightforward; not only are there dozens of EV battery types, but the recycling technologies available—such as mechanical recycling, pyrometallurgy, and hydrometallurgy—differ widely in terms of efficiency, environmental footprint, and cost. The decision to recycle or repurpose a battery depends on several factors, such as degradation level and remaining capacity of the used battery. Coordinating material transformation and transportation within this battery supply chain network, while minimizing environmental impact and cost, and enhancing the resiliency of the supply chain to ensure energy security, is challenging.

Overall, there is a pressing need to develop a holistic strategy to address the above issues for (i) the design and operations management of battery swapping stations, (ii) the pricing of the swapping services, and (iii) the managed repurpose and recycle of retired EV batteries, to minimize operating costs and emissions while maintaining the designed service levels.

3.2. Industrial Collaborations and Project Objectives

We collaborate with CATL and Nio on the operations management of battery swapping stations. Specifically, we work with CATL on the process optimization of repurpose and recycle of retired batteries, which enables us to address the last operational challenges listed in this proposal. To address the first three challenges, it is important to first understand EV owners' charging and battery swapping behavior. We have obtained access to detailed data on customers' charging behaviors from two charging stations and collected data on the time-of-day dependent demand for battery swapping in about 150 swapping stations. Next, we list the study objectives and the tasks that we plan to carry out in this proposal.

Objective 1: Capacity Design and Smart Charging of Battery Swapping Stations

Battery swapping is a relatively new business model. The capacity design of BSS needs to balance the requirement on service levels and the upfront investment cost. Moreover, the design of BSS by Nio and CATL has integrated renewable energy, especially rooftop solar energy, to promote ESG and also minimize the electricity tariffs. The stochastic patterns of the swapping demand and

renewable generations increase the complexity in making the capacity design and battery charging decisions. We plan to address those challenges through the following research tasks.

Task 1.1. Investigate the Structure of the Optimal Battery Charging Policy. We aim to study the optimal battery charging policy using a dynamic model that tracks the state of charge for each battery in the BSS. This model will account for key factors such as the stochastic nature of demand, the uncertainty of renewable energy generation, time-varying electricity prices, and the peak pricing scheme (with or without demand charges). The problem is inherently a high-dimensional decision-making challenge. Therefore, the development of efficient and effective heuristic algorithms is crucial for practical implementation. To address this, we will first investigate the structural properties of the optimal policy, with the hope of leveraging these properties to reduce computational complexity. Our initial analysis indicates that the feasible region of the decision problem does not form a lattice structure which complicates the theoretical analysis. We propose to construct a change-of-variable transformation for the original problem to address this issue. We will characterize the structure of the optimal battery charging policy, assuming that charging a battery takes one time period, to address the challenge and derive managerial insights.

Task 1.2. Structure-Informed Deep Learning Solution Method. For the results to be practically implementable, we will extend our model from single-period charging to multi-period charging. However, the system dimension grows exponentially fast and numerical evaluation of system performance becomes challenging, even for a small number of batteries under a stationary demand process. We plan to adopt deep reinforcement learning methods which use deep neural networks for value function approximation. Particularly, we will leverage the structural properties of the optimal charging policy from [Task 1.1](#) to improve the computational efficiency of our methods.

Task 1.3. Optimal Capacity Design of Battery Swapping Stations. The biggest obstacle to scaling up battery swapping services is the substantial upfront cost. As a result, designing the capacity of BSS optimally is crucial to ensure cost efficiency and operational effectiveness. We will develop a search-based optimization algorithm to determine the required capacity for renewable energy sources, e.g., solar panels, and the optimal number of spare batteries for any given service level. The solution method proposed in [Task 1.2](#) will serve as a subroutine in the optimization process.

Objective 2: Pricing Strategies of Battery Swapping Services

Pricing battery swapping services is a critical factor in ensuring the financial viability of battery swapping stations while maintaining affordability and fairness for users. Nio adjusted the pricing model for its battery swapping service in September 2024, from Per-Swap Pricing (charging a fixed fee per swap) to Mileage-Based Pricing (charging a fee based on added miles plus a service fee). However, the impacts of the change in pricing model on user adoption, revenue generation, and the overall competitiveness of this service compared to other charging solutions remain unclear.

Task 2.1. Impact of Pricing Schemes on Consumer Behavior, Swapping Station Operations, and the Environment. We will first develop a game-theoretic model to analyze the effects of two pricing schemes (per-swap and mileage-based pricing) on customers' energy replenishment behavior. Our model will consider the range anxiety, travel cost, and waiting cost (at a congested BSS) of customers. We will examine how pricing schemes influence customers' decisions on choosing between charging or battery swapping, which ultimately impacts EV adoption rate. The range anxiety of customers and pricing schemes will influence the timing of the swap decisions and the state of charge at swapping, which, in turn, endogenize the charging schedules of discharged batteries, affect congestion levels, and ultimately impact the profitability of BSS. Furthermore, the pricing schemes may also have an impact on the required number of spare batteries at the BSS to maintain the same service level. Hence, the pricing schemes affect the environments.

Task 2.2. Fair Pricing for Battery Swapping Services. In addition to choosing between per-swap pricing and mileage-based pricing, an emerging challenge for battery swapping stations is determining how to fairly allocate the total cost of providing swapping services to customers through pricing. Fair pricing for battery swapping services should account for the time-varying electricity prices and demand charges on operational costs. Customers who schedule swaps during off-peak hours, when electricity rates are lower, help reduce these expenses. Similarly, customers who offer flexibility in swap timing enable BSS to avoid peak power usage and minimize demand charges. We will build models to account for both factors in the pricing decisions to ensure fairness.

Objective 3: Optimization of Battery Repurpose and Recycling Operations

The repurpose and recycling of EV batteries are important. With the development of battery swapping services, there will be more used batteries. Hence, it is critical to manage the retired batteries effectively to minimize the environmental impacts.

Task 3.1. Optimization of Battery Recycling Infrastructures. We aim to develop a supply chain optimization framework that integrates techno-economic analysis (TEA) and life-cycle analysis (LCA) to assess the economic viability and environmental benefits of battery recycling infrastructures. This framework will involve multiple stakeholders, including used battery suppliers (e.g., swapping stations), recycling technology providers, logistics services, inventory managers, and buyers of recycled materials. It will coordinate product transformation and transportation among these entities to minimize both costs and environmental impacts. The model will be formulated as a multiperiod mixed-integer linear programming problem, where the sizes, locations, and number of facilities, as well as their operations, will be optimized simultaneously.

Task 3.2. Resilience-Enhancing Strategies for Battery Supply Chain. The battery supply chain is vulnerable due to its dependence on critical materials like lithium and cobalt, which are often sourced from regions with geopolitical risks. In this task, we aim to integrate battery recycling into the existing manufacturing supply chain and leverage stochastic optimization to design flexible, adaptive supply chains that can respond to disruptions and other uncertainties (e.g., fluctuations in used battery supplies and new battery demands). Building on the model developed in [Task 3.1](#), we will incorporate modular facility designs and alternative production methods (e.g., lithium-sulfur batteries) to enable rapid adaptation to changing conditions. Additionally, we will incorporate measures such as auxiliary equipment and increased transportation capacity to ensure a continuous supply and mitigate bottlenecks. The goal is to assess the economic costs of these resilience-enhancing strategies and provide policy insights on how to balance resilience, cost, and sustainability in the battery supply chain.

Objective 4: Understand the Charging and Swapping Behaviors of EV Owners

Understanding customer charging and swapping behavior is essential for addressing the operational challenges of battery swapping stations, detailed in Objectives 1–3. We will perform an in-depth empirical analysis of the data obtained from two charging stations and approximately 150 swapping stations, capturing time-of-day-dependent demand patterns, customer preferences, and usage behaviors.

Task 4.1. Empirical Analysis of Customer Charging and Swapping Behavior. By analyzing this rich dataset, we aim to uncover key insights into factors influencing customers' decisions, such as range anxiety, waiting times, pricing schemes, and convenience. These insights will inform the development of accurate demand forecasting models, which are critical for designing the capacity of BSS, optimizing smart charging policies, and evaluating pricing strategies. Furthermore, the analysis will reveal behavioral variations across different regions and customer segments, enabling the customization of solutions to maximize efficiency and user satisfaction. By grounding our theoretical models in empirical evidence, this task ensures the practical relevance and effectiveness of our proposed solutions for the EV ecosystem.

4. Collaboration Plan

The team members have co-supervised PhD students and collaborated on joint projects and publications. We have also established partnerships with leading companies in the battery swapping industry. The PC and some team members visited Nio and CATL in 2024 to strengthen collaboration. In this project, the PC will oversee task execution, monitor progress, and lead the overall effort. The allocation of effort across the research tasks is as follows: Yunjian Xu (Co-PI), an expert in optimal control and deep reinforcement learning, will lead Tasks 1.1, 1.2 and 1.3 with Zhankun Sun (PC) and Liao Wang (Collaborator, expert in stochastic optimal control); Wei Zhang (Co-PI, expert in game theory) will lead Task 2.1 with Zhankun Sun (PC); Co-PIs, Pengfei Guo and Jianfu Wang, both experts in behavioral modeling, will lead Task 2.2; Co-PIs, Jiaze Ma (expert in waste supply chain management) and Frank Chen (expert in inventory control and supply chain management) will lead Tasks 3.1 and 3.2; Haipeng Shen (Co-PI, expert in statistics and data analysis) will lead Task 4.1 with Zhankun Sun (PC) and Wenhui Zhou (Collaborator).